

The Wavepool Hypothesis III: “The Mechanics of Emergence”

From Pre-Geometric Potential to Patchy Early Structure Formation

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Introduction

The question of what precedes the observable universe has occupied thinkers for millennia. Aristotle, in his *Metaphysics*, described reality as arising from the actualization of potentiality (*dynamis* to *energeia*), where an underlying substrate holds the capacity for all change and form without itself being a formed thing. This ancient insight finds echoes in modern foundational physics: Max Tegmark’s Mathematical Universe Hypothesis, Stéphane Lupasco’s Logic in Reality, and recent pre-geometric frameworks all describe a pre-spacetime substrate of pure relational potential from which classical geometry, spacetime, and matter emerge.

This paper is the third and final installment in the Wavepool Hypothesis trilogy. Part 1 establishes the three irreducible fundamentals — Mathematics, Vibration, and Entropy. Part 2 develops the full Wavepool cosmology as a cold, patchy “Big Bang” arising directly from those fundamentals.

The present work details the precise mechanics of the Emergence Event (E) and the subsequent Turbulent Foam Dynamics (TFD) regime, grounding the entire series in a minimal, simulatable dynamical system.

1. Pre-Geometric Potential Space – The Three Fundamentals

Prior to the Emergence Event, reality exists as a pre-geometric relational substrate — a realm of pure potential that has not yet manifested as classical spacetime, matter, or forces. This Potential Space is governed solely by the three fundamentals in their un-manifested form:

- *Mathematics* — the pure relational structure of all possible configurations and entanglements.
- *Vibration* — oscillatory dynamics, latent as a high/low bi-pole.
- *Entropy* — intrinsic directional irreversibility, latent as a primitive bias.

These remain un-actualized until the resonant trigger at E.

2. The Emergence Event (E) – Activation of the Vibrational Pump

The Emergence Event is a sharp, resonant actualization in which the three fundamentals interact under maximal relational tension. A tiny but non-zero primitive entropy bias $\delta > 0$ activates the universal vibrational pump, splitting the latent vibrational mode into low- and high-energy poles and triggering an instability that converts pure potential into a localized classical pocket universe.

This process occurs on a timescale of order 10–40 s (near-Planck) to 10–30 s and is the precise dynamical analog of the Mexican-hat spontaneous symmetry breaking discussed in quantum field theory, now realized as an emergent instability from the pump cycle.

3. Minimal Dynamical Toy Model

To make the Emergence Event and subsequent evolution quantitatively tractable, we introduce a minimal coupled dynamical system that captures the core Wavepool ontology. The model consists of two energy reservoirs and one spatial vibrational field.

Variables

- $EL(t)$: low-vibration background energy (pervasive, non-localized)
- $EH(t)$: high-vibration localized energy (matter-like structures)
- $\phi(x,t)$: scalar vibrational field encoding structure formation (1D for minimal implementation)
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Energy Cycle (The Universal Pump)

The backbone of the model is the biased transfer cycle:

$$dtdEH = \Gamma LHEL - \Gamma HLEH$$

$$dtdEL = -\Gamma LHEL + \Gamma HLEH$$

with the entropy bias implemented as

$$\Gamma LH = \Gamma(1+\delta), \Gamma HL = \Gamma(1-\delta), 0 < \delta \ll 1.$$

Total energy $E_{\text{tot}} = E_L + E_H$ is conserved. At equilibrium the natural matter/background ratio emerges analytically:

$$E_L / E_H = (1 - \delta) / (1 + \delta).$$

This bias-driven steady state is a direct consequence of the primitive directional irreversibility (Postulate 3 of WP1) and requires no external fine-tuning.

Expansion

Cosmic expansion is driven solely by the low-vibration background:

$$\dot{a} = H = \alpha E_L,$$

where α is a coupling constant. No separate dark-energy field is required.

Vibrational Field Dynamics (Structure Formation)

The spatial field obeys a reaction-diffusion-type equation:

$$\partial_t \phi = D \nabla^2 \phi + \lambda \phi - g \phi^3 + \eta(x, t),$$

where D is diffusion, λ is linear growth from the background, $-g \phi^3$ provides saturation, and $\eta(x, t)$ is initial quantum-foam noise.

Coupling

$$E_H \propto \int \phi^2 dx, \lambda = \lambda_0 + \beta E_L, \Gamma_L H = \Gamma(1 + \delta)(1 + \epsilon \langle \phi^2 \rangle).$$

Initial conditions at $t=0$: $E_H \approx 0$, $E_L = E_0$, $\phi(x, 0) = \text{tiny noise}$. When $\lambda > 0$, fluctuations grow into localized clumps — exactly the patchy structures observed in the TFD regime.

This toy model can be simulated immediately in Python (1D grid, < 200 lines). It produces: (i) stable energy cycling with the predicted ratio, (ii) expansion driven by E_L , (iii) self-organized patchy clumps in ϕ , and (iv) self-regulation preventing runaway growth.

4. Immediate Post-E Phase – Turbulent Foam Dynamics (TFD) Regime

The pump activation at E populates the high-vibration pole and launches the TFD regime. The quantum foam consists of rapidly fluctuating fermionic components (quarks, leptons) riding on the

vibrational field ϕ . Localized peaks in ϕ generate regions of extreme field strength, enabling Schwinger-like pair production (directly analogous to recent STAR Collaboration observations of vacuum pair creation).

Gravitational attraction arises naturally as curvature sourced by the high-vibration clumps (EH), while the low-vibration background EL drives continuous expansion. The result is a convective, patchy environment lasting $\sim 380,000$ years until recombination. The model's instability directly reproduces the transient high-energy pockets required for early structure seeding.

5. Post-Fracture Evolution and Early Structure Formation

As the medium cools to ~ 3000 K, a temporary outer shell of cooler, denser material forms. Gravitational instabilities fracture this shell along tension lines created by the patchy ϕ -clumps. Released fragments collapse locally into the first stars, black holes, and proto-galaxies, while lower-density regions continue expanding under the EL-driven drive.

This mechanism, emergent from the toy model, naturally explains the JWST-observed low-metallicity luminous red dwarfs, overmassive early black holes, and other anomalous high-redshift structures without requiring a global hot Big Bang. The fracture also releases a stochastic background of ultra-low-frequency gravitational waves — a unique, testable signature of the Wavepool Emergence Event.

5.1 Neutrino Contribution to the Low-Energy Virtual Ocean

The low-energy virtual particle/boson ocean that drives the Wavepool dynamics naturally accommodates the neutrino sector. Virtual neutrino–antineutrino fluctuations, together with the ultra-cold Cosmic Neutrino Background (CvB), form an integral part of the vibrational medium. Because neutrinos are extremely weakly interacting and possess tiny masses, they behave almost as pure vacuum excitations at the energy scales relevant to that medium. Recent DESI galaxy-clustering analyses (2024–2026) that appear to require a slightly negative sum of neutrino masses may therefore represent an early observational signature of the negative-pressure vacuum contribution from the neutrino component within a Cold Big Bang framework — a feature that arises naturally in the Wavepool Hypothesis without invoking unphysical particle properties.

6. Balanced Dualities and the Advancing Entropic Wavefront

In the classical pocket universe the fundamentals manifest as balanced dualities:

- *Space* $\leftrightarrow$ *discrete granular structure of entropy*; *time* $\leftrightarrow$ *its directional drive (internal arrow + expansive dark energy)*.
- *Vibration* $\leftrightarrow$ *high/low bi-pole (bosonic fields $\leftrightarrow$ fermionic particulate fields)*.

An advancing entropic wavefront propagates through the 4D block, separating potential configurations ahead from stiffened classical history behind. Near the wavefront, time remains elastic (consistent with observed quantum tunneling anomalies). All dynamics are governed by the pump cycle and field equations above.

Conclusion

The Wavepool Hypothesis now rests on a minimal, simulatable dynamical system that begins with the three intangible fundamentals of WP1 and produces the patchy, cold emergence cosmology of WP2. The toy model demonstrates that a single biased vibrational pump plus a growing/saturating field is sufficient to generate energy cycling, expansion, and localized structure formation without external fields or fine-tuning.

The framework is falsifiable through:

- *patchy CMB spectral distortions (Simons Observatory, CMB-S4),*
- *ultra-low-frequency gravitational waves from E and shell fracture (LISA, tabletop experiments),*
- *continued early patchy structure (JWST, Roman).*

Future work will extend the toy model to 3D, incorporate full gravitational feedback, and generate quantitative predictions for the above signatures. For the first time, the Wavepool series offers not only a coherent philosophical ontology but a concrete, runnable dynamical engine for the emergence of our universe.

References

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